
Introduction to Supervised Learning

MSQF 536 – Machine Learning Techniques

Unit II

Supervised Learning

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2.1 What is Supervised Learning?

Core idea

Supervised learning learns a mapping from **input variables** to a **known target variable** using labelled observations.



- ▶ Every training observation contains both **features** and a **known response/label**.
- ▶ The algorithm learns the relationship between $\mathbf{x}$ and y .
- ▶ The resulting model is then used to predict y for new $\mathbf{x}$.
- ▶ The central objective is **generalization**, not memorization.

Quantitative Finance Example

Suppose we have historical information on a stock:

$$\mathbf{x}_i = (\text{past return, volatility, volume, interest rate, market return})$$

and the target is the **next-day return** y_i .

A supervised-learning model attempts to learn

$$\hat{r}_{t+1} = \hat{f}(\text{market and stock information at time } t)$$

Mathematical Framework of Supervised Learning

Let the observed training sample be

$$\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\},$$

where

$$\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})^\top$$

contains p predictor variables and y_i is the response.

The learning algorithm searches for a function

$$f : \mathbb{R}^p \rightarrow \mathcal{Y}$$

such that

$$\hat{y}_i = \hat{f}(\mathbf{x}_i)$$

provides accurate predictions.

A general empirical-risk formulation is

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n L(y_i, f(\mathbf{x}_i; \theta))$$

where L measures the discrepancy between the observed and predicted responses.

- ▶ θ = model parameters.
- ▶ L = loss function.
- ▶ $\hat{\theta}$ = parameters learned from the training data.

Quantitative Finance Interpretation

For predicting stock returns, if $y_i = r_{i,t+1}$, then a regression model may minimize squared prediction errors:

$$L(y_i, \hat{y}_i) = (y_i - \hat{y}_i)^2.$$

The fitted model therefore seeks parameters that make predicted returns close to observed future returns in the training sample.

Regression and Classification

The nature of the target variable determines the supervised-learning task.

Problem	Target variable	Quantitative-finance examples
Regression	Continuous	Predict next-day return, asset price, volatility, portfolio loss or trading volume
Binary classification	Two classes	Predict whether a stock return is positive/negative; default/non-default
Multiclass classification	Three or more classes	Classify market condition as bullish, neutral or bearish

Important observation

The same financial predictors can be used in both regression and classification. The key difference is the **target** and therefore the statistical model and loss function used.

Example

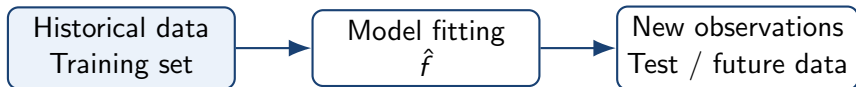
Using past returns, volatility and trading volume:

$$\text{Regression: } \hat{r}_{t+1} = f(\mathbf{x}_t)$$

$$\text{Classification: } \hat{C}_{t+1} = \begin{cases} 1, & r_{t+1} > 0, \\ 0, & r_{t+1} \leq 0. \end{cases}$$

The first predicts **how much** the return may be; the second predicts **which class** it belongs to.

Training, Prediction and Generalization



- ▶ **Training error:** error calculated using observations that were used to fit the model.
- ▶ **Test error:** error calculated using observations not used during model fitting.
- ▶ **Generalization:** ability of the learned model to perform well on unseen observations.
- ▶ A very flexible model can achieve low training error but poor out-of-sample performance.

Key statistical idea

A useful supervised-learning model must balance

good fit to observed data and good prediction for unseen data.

Quantitative Finance

Financial time series make this distinction particularly important. A model that predicts historical stock returns extremely well may fail on future market data because the model has captured historical noise rather than a stable predictive relationship.

Key Concepts and the Supervised Learning Pipeline

Essential terminology

- ▶ Observation / instance
- ▶ Feature / predictor / covariate
- ▶ Response / target / label
- ▶ Training set
- ▶ Test set
- ▶ Model
- ▶ Prediction
- ▶ Loss function
- ▶ Hyperparameter

Typical workflow

1. Define prediction problem
2. Collect labelled data
3. Pre-process data
4. Split data appropriately
5. Select a model
6. Train the model
7. Tune hyperparameters
8. Evaluate out-of-sample
9. Interpret and deploy

Features $\rightarrow$ Learning Algorithm $\rightarrow$ Fitted Model $\rightarrow$ Prediction $\rightarrow$ Evaluation

Quantitative Finance Connection

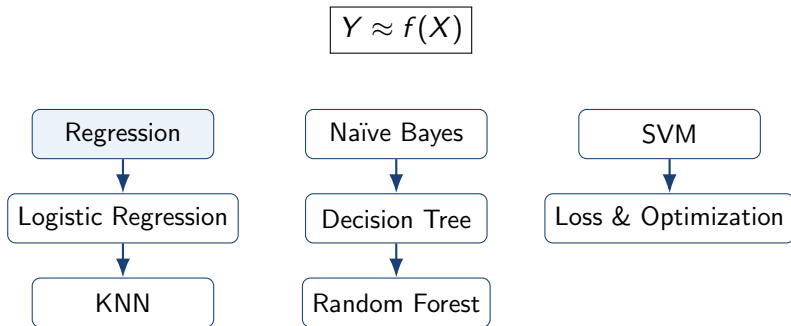
For a stock-return prediction problem:

Market data $\rightarrow$ Features $\rightarrow$ Supervised model $\rightarrow \hat{r}_{t+1} \rightarrow$ Trading / risk decision.

The final financial decision should be based on **out-of-sample** performance rather than training performance alone.

Where the Unit is Heading

The supervised-learning methods in this unit can be viewed as different ways of estimating the unknown relationship



Master-level perspective

Although the algorithms differ, they can be studied through a common framework:

Data → Model → Loss / Objective → Optimization → Prediction → Generalization

Understanding this common structure makes it easier to compare Logistic Regression, KNN, Naïve Bayes, Tree Models and SVM.

2.2 Framework of Regression

Basic idea

Regression is a supervised-learning framework in which the response variable Y is quantitative, and the objective is to learn the conditional relationship between Y and a set of predictors $\mathbf{X}$.

For observations

$$\mathcal{D} = \{(\mathbf{x}_i, y_i) : i = 1, \dots, n\},$$

the regression problem is to estimate

$$m(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}]$$

or, more generally, a function

$$Y = f(\mathbf{X}) + \varepsilon.$$

- ▶ $\mathbf{X} = (X_1, \dots, X_p)^\top$: predictor variables.
- ▶ Y : continuous response variable.
- ▶ $f(\mathbf{X})$: systematic component explained by predictors.
- ▶ ε : unexplained/random component.

Quantitative Finance Example

Let $Y = r_{t+1}$ denote the next-day return of an asset and

$$\mathbf{X}_t = (r_t, \sigma_t, V_t, r_t^m, \text{interest rate}_t).$$

A regression model attempts to estimate

$$\mathbb{E}[r_{t+1} \mid \mathbf{X}_t] = f(\mathbf{X}_t)$$

and hence predict the next-period return.

Statistical Structure of a Regression Model

A useful decomposition is

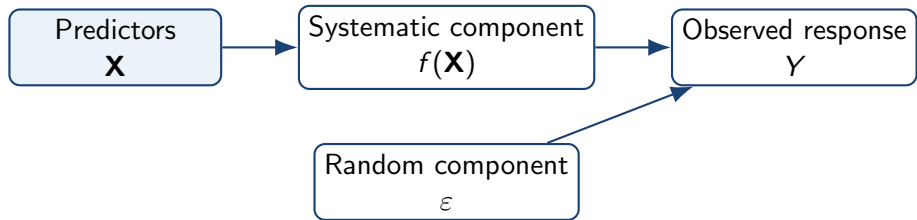
$$Y_i = f(\mathbf{X}_i) + \varepsilon_i$$

with

$$\mathbb{E}[\varepsilon_i \mid \mathbf{X}_i] = 0.$$

Thus,

$$\mathbb{E}[Y_i \mid \mathbf{X}_i] = f(\mathbf{X}_i).$$



- ▶ Regression separates the predictable structure from random variation.
- ▶ The conditional mean is the optimal prediction under squared-error loss.
- ▶ Different regression models impose different assumptions about f .

Finance Interpretation

For asset returns,

$$r_{t+1} = f(\mathbf{X}_t) + \varepsilon_{t+1}.$$

Even if the predictors contain information about future returns, ε_{t+1} represents movements that cannot be predicted from $\mathbf{X}_t$ under the chosen information set.

Parametric Regression Framework

In a parametric regression model, the functional form is specified up to a finite-dimensional parameter vector β :

$$Y_i = f(\mathbf{X}_i; \beta) + \varepsilon_i.$$

For multiple linear regression,

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip} + \varepsilon_i.$$

In matrix notation,

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon.$$

- ▶ β_0 : intercept.
- ▶ β_j : change in the conditional mean associated with X_j , holding the other predictors fixed.
- ▶ β : unknown parameter vector estimated from data.

Modeling choice

The regression framework is broader than linear regression. The function f may be linear, nonlinear, polynomial, regularized, tree-based, kernel-based, or another suitable predictive function.

Quantitative Finance Example

A simple return model may be

$$r_{t+1} = \beta_0 + \beta_1 r_t + \beta_2 \sigma_t + \beta_3 r_t^m + \varepsilon_{t+1}.$$

Here $\beta_1, \beta_2, \beta_3$ quantify the fitted relationship between current financial variables and the subsequent return.

Estimation: From Data to a Fitted Regression Model

The unknown parameters must be estimated from the training data.
For squared-error loss,

$$L(y_i, \hat{y}_i) = (y_i - \hat{y}_i)^2.$$

The empirical risk is

$$R_n(\beta) = \frac{1}{n} \sum_{i=1}^n [y_i - f(\mathbf{x}_i; \beta)]^2.$$

The fitted parameters are

$$\hat{\beta} = \arg \min_{\beta} R_n(\beta).$$

For the linear model, this gives the ordinary least-squares estimator

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

when $\mathbf{X}^\top \mathbf{X}$ is nonsingular.

- ▶ $\hat{y}_i = f(\mathbf{x}_i; \hat{\beta})$: fitted value.
- ▶ $e_i = y_i - \hat{y}_i$: residual.
- ▶ The optimization chooses parameters that minimize the total squared discrepancy between observations and fitted values.

Finance Example

If the observed next-day return is 1.2% and the fitted model predicts 0.8%, then

$$e_i = 1.2\% - 0.8\% = 0.4\%.$$

Least squares estimates the coefficients so that the aggregate squared prediction errors over the training observations are minimized.

Prediction, Risk and Generalization

A fitted regression model produces

$$\hat{y}_{\text{new}} = \hat{f}(\mathbf{x}_{\text{new}}).$$

The important distinction is between **training performance** and **out-of-sample performance**.



A useful statistical objective is the expected prediction risk

$$R(f) = \mathbb{E}[L\{Y, f(\mathbf{X})\}].$$

Since the true distribution is unknown, this risk must be approximated using validation/test data or suitable resampling procedures.

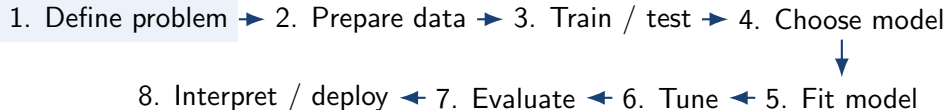
Key principle

A regression model is useful when it provides reliable **out-of-sample predictions**, not merely a small training error.

Quantitative Finance

A return-prediction model should be evaluated on observations occurring **after the training period**. Randomly mixing future observations into the training set can introduce look-ahead bias and produce an unrealistically optimistic assessment.

Regression Modeling Workflow



Regression framework in one view

Predictors → Model f → Loss → Parameter estimation → Prediction → Risk evaluation

- ▶ **Model selection:** What functional form should be used?
- ▶ **Estimation:** How should unknown parameters be learned?
- ▶ **Evaluation:** How well does the model generalize?
- ▶ **Interpretation:** What does the fitted relationship mean?

Quantitative Finance Perspective

For forecasting an asset return, the complete pipeline connects financial information $\mathbf{X}_t$ to a future response r_{t+1} , while respecting temporal ordering, avoiding information leakage, and evaluating predictive performance out-of-sample.

Framework of Regression: Key Takeaways

1. Regression models the relationship between a quantitative response Y and predictors $\mathbf{X}$.
2. The fundamental statistical representation is

$$Y = f(\mathbf{X}) + \varepsilon.$$

3. The conditional mean

$$\mathbb{E}[Y \mid \mathbf{X}]$$

is the optimal predictor under squared-error loss.

4. A regression algorithm estimates an unknown function or its parameters using training data.
5. Loss functions provide a mathematical criterion for fitting the model.
6. Prediction quality must ultimately be judged on **unseen data**.
7. Model complexity introduces the possibility of **overfitting** and poor generalization.
8. In quantitative finance, temporal ordering and information leakage are particularly important considerations.

2.3 Model: The Mathematical Representation

A **model** is a mathematical representation of the relationship between predictors $\mathbf{X}$ and a response Y .

$$Y = f(\mathbf{X}; \boldsymbol{\theta}) + \varepsilon$$

where

- ▶ $\mathbf{X} = (X_1, \dots, X_p)^\top$ = predictor variables,
- ▶ Y = response / target variable,
- ▶ $f(\cdot)$ = functional form of the model,
- ▶ $\boldsymbol{\theta}$ = unknown model parameters,
- ▶ ε = random/unexplained component.

The fitted model is obtained after learning the parameters from training data:

$$\hat{Y} = \hat{f}(\mathbf{X}; \hat{\boldsymbol{\theta}})$$

Important distinction

A **model** specifies how predictions are generated; a **learning algorithm** specifies how the model parameters are estimated from data.

Quantitative Finance Example

To predict tomorrow's stock return,

$$r_{t+1} = f(r_t, \sigma_t, V_t, r_t^m; \theta) + \varepsilon_{t+1}.$$

The predictors may include today's return r_t , volatility σ_t , trading volume V_t , and market return r_t^m .

Model Choices in Supervised Learning

Different models make different assumptions about the relationship between **X** and **Y**.

Model type	Main idea	Examples
Parametric	Fixed functional form with finite parameters	Linear regression, logistic regression
Non-parametric	Flexible functional form; complexity grows with data	KNN, kernel methods, trees
Linear	Prediction is linear in predictors / parameters	$\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$
Nonlinear	Captures nonlinear relationships	Polynomial models, trees, neural networks

Model selection

The best model is not necessarily the most complex model. The objective is to obtain an appropriate balance between **flexibility, predictive accuracy, interpretability and generalization**.

Quantitative Finance

A linear model may assume

$$r_{t+1} = \beta_0 + \beta_1 r_t + \beta_2 \sigma_t + \varepsilon_{t+1},$$

while a tree-based model can capture interactions such as

“high volatility” $\times$ “negative market return”

without explicitly specifying the interaction beforehand.

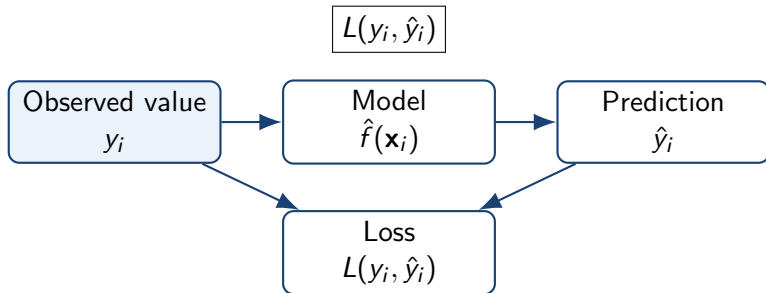
Why Do We Need a Loss Function?

For observation i , the model produces $\hat{y}_i = \hat{f}(\mathbf{x}_i)$.

The prediction error is

$$e_i = y_i - \hat{y}_i.$$

A **loss function** quantifies how undesirable a prediction error is:



- ▶ A loss function converts prediction errors into a numerical quantity.
- ▶ Different losses express different notions of “bad prediction”.
- ▶ The choice of loss influences the fitted model.

Quantitative Finance

Suppose the actual next-day return is

$$y = 0.020$$

and the model predicts

$$\hat{y} = 0.012.$$

Then the prediction error is

$$e = y - \hat{y} = 0.008.$$

A loss function determines how strongly this 0.8% prediction error contributes to model fitting.

Squared Error Loss and Mean Squared Error

The most common regression loss is the **squared error loss**:

$$L_{\text{SE}}(y, \hat{y}) = (y - \hat{y})^2$$

For n observations, the corresponding mean squared error is

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

The model-fitting problem becomes

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n [y_i - f(\mathbf{x}_i; \theta)]^2.$$

- ▶ Squaring makes all losses non-negative.
- ▶ Large errors receive disproportionately large penalties.
- ▶ The loss is differentiable, which is useful for optimization.
- ▶ Under squared-error loss, the optimal prediction is the conditional mean:

$$f^*(\mathbf{x}) = \mathbb{E}[Y \mid X = \mathbf{x}].$$

Quantitative Finance

Consider actual returns (0.02, -0.01, 0.03) and predictions (0.015, 0.00, 0.025).
The squared errors are

$$(0.005)^2, \quad (-0.01)^2, \quad (0.005)^2.$$

Hence

$$\text{MSE} = \frac{0.000025 + 0.0001 + 0.000025}{3} = 0.00005.$$

Alternative Regression Loss Functions

Squared error is not the only possible choice.

Loss	Definition	Main characteristic
MSE	$(y - \hat{y})^2$	Strongly penalizes large errors
MAE	$ y - \hat{y} $	More robust to outliers
Huber	Quadratic for small errors, linear for large errors	Compromise between MSE and MAE

For MAE,

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Under absolute-error loss, the optimal prediction is related to the **conditional median**, rather than the conditional mean.

Why does the choice matter?

Different loss functions imply different attitudes toward prediction errors and therefore can lead to different fitted models.

Quantitative Finance

Financial returns can contain extreme observations caused by market crashes or major announcements.

If a few extreme returns dominate MSE, an analyst may consider MAE or Huber loss when the goal is to obtain a more robust estimate of the typical predictive relationship.

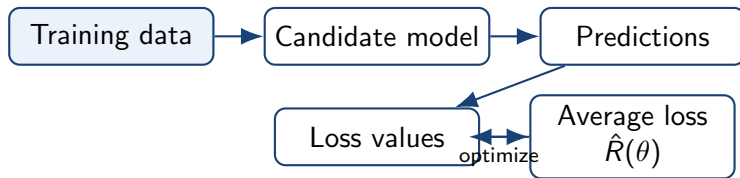
Loss Function → Empirical Risk Minimization

For a training dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$,
the average loss is called the **empirical risk**:

$$\hat{R}(\theta) = \frac{1}{n} \sum_{i=1}^n L(y_i, f(\mathbf{x}_i; \theta)).$$

The learning algorithm solves

$$\hat{\theta} = \arg \min_{\theta} \hat{R}(\theta).$$



Central machine-learning idea

Choose model parameters that minimize empirical prediction loss.

Quantitative Finance

For return prediction, the training objective under MSE is

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{n} \sum_{t=1}^n [r_{t+1} - f(\mathbf{x}_t; \theta)]^2.$$

The algorithm searches for parameters producing the smallest average squared prediction error over the training period.

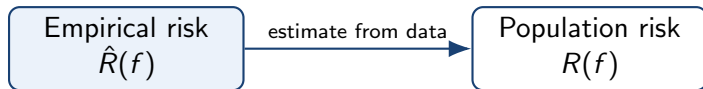
Empirical Risk vs Generalization Risk

The quantity we can calculate from training data is

$$\hat{R}(f) = \frac{1}{n} \sum_{i=1}^n L(y_i, f(\mathbf{x}_i)).$$

But the quantity we ultimately care about is the **expected prediction risk**:

$$R(f) = \mathbb{E}_{(X,Y)} [L(Y, f(X))].$$



- ▶ Minimizing training loss alone can produce **overfitting**.
- ▶ The true objective is good performance on future observations.
- ▶ Validation, cross-validation and test sets help estimate out-of-sample performance.

Quantitative Finance

A model may achieve extremely low MSE during 2015–2024 but perform poorly during 2025.

Therefore,

low training MSE $\nRightarrow$ good future trading prediction.

Temporal validation is particularly important because financial observations are ordered in time.

Loss Function with Regularization

Minimizing prediction loss alone may produce an overly complex model.

A general regularized objective is

$$\hat{\theta} = \arg \min_{\theta} \left[\frac{1}{n} \sum_{i=1}^n L(y_i, f(\mathbf{x}_i; \theta)) + \lambda \Omega(\theta) \right].$$

Here,

- ▶ L = data-fitting loss,
- ▶ $\Omega(\theta)$ = complexity penalty,
- ▶ $\lambda \geq 0$ = regularization strength.

Two important examples are:

$$\text{Ridge: } \Omega(\beta) = \sum_{j=1}^p \beta_j^2, \text{ and LASSO: } \Omega(\beta) = \sum_{j=1}^p |\beta_j|.$$

Interpretation

The optimization now balances two objectives:

fit the data + control model complexity.

Quantitative Finance

Suppose hundreds of financial indicators are available to predict returns. An unregularized model may assign extreme coefficients to noise-driven predictors. Regularization can shrink coefficients and potentially improve out-of-sample stability.

Model + Loss Function: Complete Framework

The complete supervised regression framework can be summarized as

Data: $(\mathbf{x}_i, y_i)$

Model: $\hat{y}_i = f(\mathbf{x}_i; \theta)$

Error: $e_i = y_i - \hat{y}_i$

Loss: $L(y_i, \hat{y}_i)$

Risk: $\hat{R}(\theta) = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{y}_i)$

Learning: $\hat{\theta} = \arg \min_{\theta} \hat{R}(\theta).$



Master-level takeaway

A machine-learning model is not defined only by its functional form. The **model, loss function, optimization criterion and evaluation strategy together determine what the learning procedure actually learns.**

Quantitative Finance

For next-day return forecasting, the choice between MSE, MAE or a regularized objective changes what the fitted model considers an “acceptable” prediction error and can therefore change the resulting financial predictions.